

Task 4: Responsible AI & Model Interpretation

Internspark AI Internship

1. GitHub Repository

<https://github.com/zom1xo/responsible-ai-churn>

2. Summary

Conducted model interpretability and fairness audit on the churn prediction model from Task 1. Applied SHAP for global and local explanations, LIME for instance-level interpretability. Evaluated bias across gender, senior citizen status, and contract type using FPR and FNR. Found no significant gender disparity (FPR within 0.01, FNR within 0.02). Senior recall was slightly higher than non-seniors (0.561 vs 0.489). Primary limitation: low overall recall (0.508). Recommended SMOTE/class weighting and post-deployment fairness monitoring.

3. Key Findings

- SHAP, LIME, and built-in importance all identified tenure, contract type, and monthly charges as top predictors.
- No significant bias detected across gender or senior citizen groups.
- Model misses ~49% of churners due to class imbalance.

4. Tools

Python, Scikit-learn, SHAP, LIME, Pandas, Matplotlib, Google Colab